

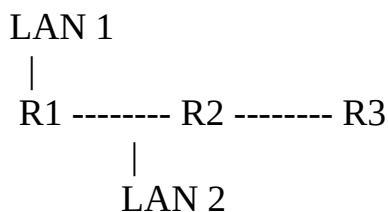
Q-Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.
Hint: Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

Routing Information Protocol (RIP) is a dynamic routing protocol used by routers to exchange information about available networks. RIP automatically updates routing tables when a network path becomes unavailable.

In this experiment, a network failure was simulated by shutting down one router interface in a RIP-enabled topology. The routing tables of the neighboring routers were monitored using the show ip route command to observe how RIP detected the failure and removed the unreachable route.

Network Topology

The experiment uses a RIP-enabled network consisting of multiple routers and connected LANs.



RIP is configured on the routers so that they learn remote networks dynamically.

Required Commands

```
show ip interface brief
show ip route
show ip route rip
show ip protocols
configure terminal
interface gigabitEthernet 0/0
shutdown
no shutdown
```

Initial Configuration Verification

Before simulating the failure, the routing table was checked on the routers.

```
R2# show ip route
```

Routes learned through RIP are identified by the letter R.

Example:

R 192.168.1.0/24 [120/1] via 10.0.0.1

Procedure

Step 1: Check the Router Interfaces

On Router R1, the interfaces were checked using:

```
R1# show ip interface brief
```

The interface connecting R1 to the rest of the RIP topology was identified.

Step 2: Shut Down the Interface

The selected interface was administratively disabled:

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# shutdown
```

This simulated a network-link failure.

Step 3: Monitor the Neighboring Router

Immediately after shutting down the interface, Router R2 was monitored repeatedly using:

```
R2# show ip route
```

Initially, R2 could still display the RIP route because the routing information had not yet timed out.

Step 4: Observe Route Removal

The show ip route command was repeatedly executed until the unreachable RIP route disappeared from R2's routing table.

The time between shutting down the interface and removal of the route was recorded.

7. Observation

Event	Observation
Initial state	RIP routes were present
Interface shutdown	R1 interface became administratively down
Immediately after failure	R2 could still show the RIP route
During convergence	RIP updated its routing information

After convergence	Unreachable route was removed
Approximate convergence time	About 180–240 seconds under default RIP timers

8. RIP Convergence

RIP uses timers to determine whether a route is still reachable.

Timer	Default Value	Purpose
Update Timer	30 seconds	RIP sends routing updates
Invalid Timer	180 seconds	Route is considered invalid if updates are not received
Hold-down Timer	180 seconds	Helps prevent unstable route changes
Flush Timer	240 seconds	Route is removed from the routing table

RIP is therefore relatively slow at detecting and removing failed routes compared with many modern routing protocols.

9. Verification After Failure

The routing table was checked using:

```
R2# show ip route
```

The previously learned RIP route was no longer present after convergence.

RIP configuration and timers can also be examined using:

```
R2# show ip protocols
```

RIP-specific routes can be displayed using:

```
R2# show ip route rip
```

10. Restoring the Network

After completing the experiment, the interface was restored using:

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no shutdown
```

The routers then exchanged RIP updates again, and the previously unavailable route was eventually learned again.

The experiment successfully demonstrated RIP convergence.

When the router interface was shut down, the connected path became unavailable. The neighboring router did not immediately remove the route from its routing table. RIP required time to detect the failure through its routing-update and routing timers.

After the convergence period, the unreachable route was removed from the routing table.

Observed/expected convergence time: approximately 3–4 minutes under default RIP timing, depending on the exact failure timing and simulation environment.